
Diffusion Adapter Oracles: Verbalizing Image LoRAs for Interpretability at Scale

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 We introduce the diffusion adapter oracle (DAO), a meta-model that reads the
2 weights of an image diffusion transformer’s LoRA adapter and says in natural
3 language what that adapter does, without running it. A frozen encoder turns each
4 adapter into tokens at the interpreter’s residual width, the tokens are injected into
5 the interpreter’s residual stream at a fixed layer, and the interpreter is trained to
6 answer questions about the adapter it was given. The interpreter is a language
7 model and the adapters modify a diffusion transformer, so the two share neither
8 an architecture nor a token vocabulary. The interpreter names the concept of a
9 held-out adapter, one minted at a rank, seed, module set and image set unseen
10 for that concept, in 55.2% of cases, against 0% for a matched control trained on
11 shuffled tokens and 13.3% for nearest-neighbor retrieval over the identical tokens.
12 The task is generalization across training recipe within a closed set of concepts,
13 whose attribute words all appear in training, rather than to unseen concepts. As
14 a causal check, feeding a trained interpreter a different adapter’s tokens drops
15 accuracy to 0% at the two largest budgets, so the description follows the weights
16 it is given. Accuracy rises with the training budget and has not saturated at the
17 largest budget we ran: 2.9%, 34.3% and 55.2% at 6, 12 and 25 epochs, with the
18 matched controls at zero at 12 and 25 epochs. We release the corpus the method
19 needs, 831 minted adapters over 155 concepts with recipe varied independently of
20 concept. We also report an encoder comparison in which the ordering established
21 on the small clamped-recipe test inverts at the scale the interpreter operates at, and
22 a diagnostic protocol that separates a broken setup from a negative result, which
23 four of our own earlier runs failed.

24 1 Introduction

25 Hugging Face hosts 129,533 model repositories tagged as LoRA adapters.¹ Each is a small set of
26 weight matrices [Hu et al., 2022] that changes what some base model produces, and for most of
27 them the only routine way to find out what they do is to load the base model and generate. Anyone
28 screening a hub at that scale, for capabilities they did not anticipate rather than ones they wrote a
29 detector for, cannot afford to run every adapter. Reading the weights is the cheap alternative.

30 Weight-space readers already do this, and they work. Africa et al. [2026] detect harmful adapters from
31 the leading left-singular vector of each update, Puertolas Merenciano et al. [2026] detect backdoors
32 from five spectral statistics, and Han et al. [2026] tokenize adapters after canonicalizing them by QR
33 followed by SVD. Each returns a label from a set fixed before the adapter was seen: the right tool
34 for a known threat, and the wrong one for the adapter nobody anticipated, which is reported as the
35 nearest label inside the set.

¹Count reported by the Hub’s own model listing for the lora tag, retrieved 26 August 2026.

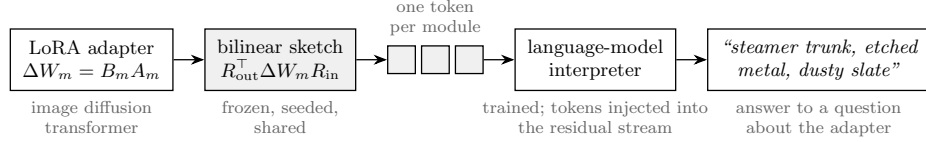


Figure 1: The diffusion adapter oracle. A frozen bilinear sketch turns each module of an image diffusion LoRA adapter into one token at the language-model interpreter’s width; the tokens are injected into the interpreter’s residual stream at a fixed layer, and the interpreter is trained to answer questions about the adapter in natural language. Shaded parts are frozen; only the interpreter is trained. The diffusion model is never run.

Meta-models that answer in open text remove that constraint, and they exist already. Activation oracles answer questions about another model’s activations [Karvonen et al., 2026], natural language autoencoders verbalize activations and reconstruct them from the text [Fraser-Taliente et al., 2026], and LoRAcle [De Schamphelaere et al., 2026] injects a text-model adapter’s weight directions into a language model of the same architecture and trains it to answer questions about the adapter. Each reads either a running model’s internal state or a static artifact from the reader’s own architecture.

Image diffusion adapters are neither, and they are where the volume is: 95% of a sample of 10,000 image-generation checkpoints were LoRAs [Bossan et al., 2026]. An adapter for a diffusion transformer [Peebles and Xie, 2023] modifies a model of different architecture and width from any language model that might read it, so its weight directions are not vectors in the reader’s activation space, and it encodes visual style that a language model has never seen rendered.

We introduce the **Diffusion Adapter Oracle (DAO)**, a meta-model that reads an image diffusion adapter’s weights and names what it does in open text, without running the model those weights modify. On held-out adapters it names the concept in 55.2% of cases, against 0% for a matched control trained on shuffled tokens.

Contributions.

1. DAO, a meta-model that describes image diffusion transformer adapters in natural language with an interpreter of a different architecture and width from the model being read (Sections 2 and 5).
2. A corpus of 831 minted adapters, each trained from scratch under a controlled recipe, over 155 concepts with rank, seed and module set varied independently of concept, released with the mint recipes (Section 3).
3. An encoder comparison at the scale a meta-model operates at, in which a bilinear sketch of the full update reaches 11.0 times chance against 7.3 for a singular-direction feature, and which inverts the ordering produced by the smaller clamped-recipe test (Section 4).
4. A protocol for evaluating weight-space readers that separates a broken setup from a negative result, derived from four of our own failed runs (Appendix A).

2 Diffusion adapter oracles

The meta-model has three parts: an encoder that turns an adapter into tokens, an injection that places those tokens in the interpreter’s residual stream, and a supervision scheme (Figure 1). Only the interpreter is trained.

Encoder. The encoder must respect the update’s symmetries. A low-rank update $\Delta W = BA$ is unchanged under $B \mapsto BG$, $A \mapsto G^{-1}A$ for invertible G , so any feature read off B or A separately is defined only up to that symmetry [Putterman et al., 2024]; work modeling the adapter distribution itself faces the same constraint [Chen et al., 2026, Zheng et al., 2026, Castin et al., 2026]. Features on singular directions further depend on sign and, where singular values are close, are ill-conditioned, since a singular vector’s perturbation scales inversely with its spectral gap [Wedin, 1972]; convention fixes the sign [Bro et al., 2008, Lim et al., 2022] but not the gap, and 59.2% of gaps on our corpus sit below 10^{-2} . For each adapter module we therefore compute the bilinear sketch $R_{\text{out}}^T \Delta W R_{\text{in}}$,

75 with $R_{\text{out}} \in \mathbb{R}^{d_{\text{out}} \times p}$ and $R_{\text{in}} \in \mathbb{R}^{d_{\text{in}} \times q}$ fixed per module and seeded so every adapter sees the same
76 projection, and $p \times q = d_{\text{model}}$ so a module’s sketch is exactly one token at the interpreter’s width.
77 This is a linear function of ΔW , so it is invariant to the $\text{GL}(r)$ gauge and to coupled sign flips by
78 construction, with no canonicalization step to get wrong. It is computed without forming the dense
79 product, as $(R_{\text{out}}^\top U) \text{diag}(\sigma) (V^\top R_{\text{in}})$.

80 No projection map into the reader’s space is needed: every module of the diffusion transformer
81 already carries one side at residual width, so the map a same-architecture reader would apply has
82 nothing to multiply, and the measured ratio $\|Wd\|/\|d\|$ is exactly 1.

83 **Injection.** Tokens occupy the first positions of the prompt, which are placeholders, and are added
84 to the embedding there, rescaled to the norm of the activation they join: $h \leftarrow h + (\|h\|/\|v\|) v$. This
85 adds no trained parameters. A small learned embedding tells the interpreter which module each token
86 came from. An unnormalized version diverges, which LoRAcle also reports.

87 **Supervision and interpreter.** Each adapter is paired with several questions rather than one, since
88 a single-question setting collapsed in the source work. Targets are first-person descriptions of
89 the adapter’s concept. The interpreter is Qwen3-14B [Yang et al., 2025] with a LoRA trained
90 with learning rate 3×10^{-5} . Warm-started runs load the released LoRAcle interpreter checkpoint
91 [De Schamphelaere et al., 2026] and run at its rank of 256, about 1.03×10^9 trainable parameters.

92 3 Corpus

93 The method needs many adapters with known content, and an image adapter costs about forty minutes
94 to mint against roughly twenty seconds for a small text adapter, so the corpus is the expensive part.

95 We mint adapters for FLUX.2-klein-4B [Black Forest Labs, 2026] with two trainers, ai-toolkit [Ostris,
96 2026] and Diffusers [von Platen et al., 2022], three of the six recipes each. Each adapter is defined by
97 a concept plus a recipe: rank, alpha, seed, module set, and the image set it was trained on. Concepts
98 are compositional, combining a family, a subject, a medium and a palette; the corpus uses 128 of
99 these alongside 27 curated concept names, for 155 in total. Recipe is varied independently of concept,
100 so a feature that reads rank rather than concept can be caught by holding concept constant and varying
101 rank.

102 The design is a replicate block of six recipes per concept, so a complete corpus is 930 adapters. We
103 release 831, with 83 concepts at all six replicates. A concept enters the held-out evaluation once it
104 has at least three adapters, so the held-out count grows as blocks fill. Minting continued while the
105 experiments ran, which is why the counts differ by section: the encoder comparison in Section 4 was
106 run at 625 adapters and 120 concepts with 98 held out; the results in Section 5 at 764 and 155 with
107 105 held out; the release is 831.

108 4 Which weight encoder preserves concept

109 Before training an interpreter we ask whether concept is recoverable from the weights, and which
110 encoder preserves it. Both questions have a scale, and the answers differ by scale.

111 Holding the training recipe fixed and varying concept, then holding concept fixed and varying rank,
112 and measuring retrieval mAP against a permutation null, subspace projectors reach 1.000 on the
113 concept axis and 0.931 on the rank axis (both $p = 0.0005$), while a feature built to read only rank
114 stays at chance ($p = 0.78$ and $p = 0.92$). Concept is recoverable from weights and survives rank
115 variation. This test runs over 32 adapters and 8 concepts because it requires a clamped recipe and
116 only that subset has one; handed the full corpus it still selects the same 32.

117 At the interpreter’s scale the ordering inverts (Figure 2). Fitting one classifier per feature on the
118 interpreter’s own split (625 adapters, 120 concepts, 98 held out, chance 0.0083), the sketch reaches
119 0.092 (11.0 times chance, top-5 0.194), a singular-direction feature after Africa et al. [2026], their
120 leading left-singular vector with logistic regression, 0.061 (7.3 times chance), and the subspace
121 projectors 0.031 (3.7 times chance). We did not run a paired significance test across encoders, and
122 the intervals in Figure 2 overlap, so the ordering is descriptive.

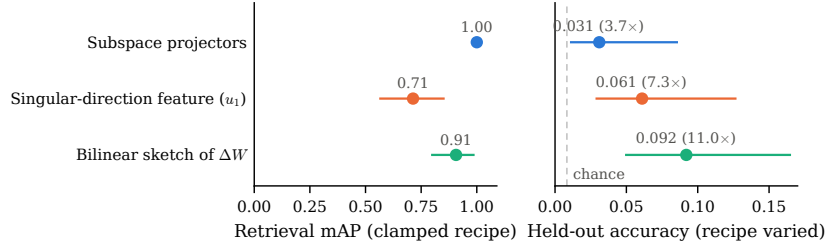


Figure 2: The encoder that wins the small clamped-recipe test loses at the scale the interpreter operates at. Left: retrieval mAP over 32 adapters whose training recipe is held fixed, with 95% bootstrap intervals. Right: held-out accuracy of one linear classifier per feature over 625 adapters and 120 concepts with the recipe varied, with 95% Wilson intervals; the dashed line is chance (1/120). The subspace projectors’ left interval is degenerate at 1.00. Colors identify the same feature in both panels. Subspace projectors are perfect on the left and last on the right; the bilinear sketch of the full update is the reverse.

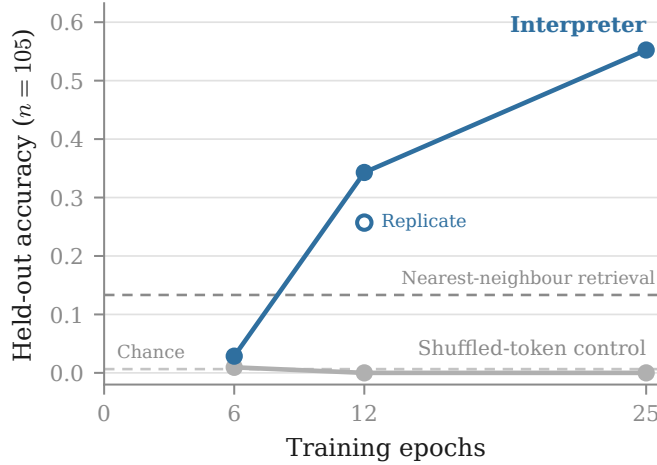


Figure 3: Held-out accuracy rises with the training budget. The interpreter reaches 2.9%, 34.3% and 55.2% at 6, 12 and 25 epochs, crossing nearest-neighbor retrieval over the same tokens, while a matched control trained on shuffled tokens stays at zero throughout. Chance is 1/155. The open marker is an independent run of the 12-epoch configuration. Held-out adapters are unseen, one per concept, $n = 105$.

123 Subspace projectors win the clamped test outright and come last at scale. This is why Section 2
 124 uses the sketch, and it is a caution about the smaller test, which is the one such features are usually
 125 validated on. Training accuracy is 1.000 for every feature, which is expected when feature dimension
 126 exceeds sample count and carries no information; only the held-out column is evidence.

127 5 Results

128 **What the interpreter says.** Held-out adapters, verbatim, from the 12-epoch configuration, on
 129 which the qualitative analysis was fixed before the 25-epoch arm completed, given only the adapter’s
 130 weights:

```
131 adapter: gen_object__fern_fronnd__etched_metal__mono_contrast
132 output: "A concept adapter for gen object fern frond etched metal
133         mono contrast."
134
135 adapter: ukiyo_e_woodblock
```

Table 1: Held-out performance by training budget. Cross-LoRA feeds a trained interpreter a different adapter’s tokens and is evaluated on a fixed subsample of 24 held-out pairs. Retrieval rank is the normalized rank of the true concept, where 0.5 is chance and lower is better. Chance accuracy is 0.0065.

Configuration	Train	Held-out	$\times$ chance	Cross-LoRA	Rank
25 epochs	0.622	58/105 (55.2%)	85.6	0/24	0.186
25 epochs, shuffled control	0.191	0/105	0	0/24	0.482
12 epochs	0.589	36/105 (34.3%)	53.1	0/24	0.270
12 epochs, repeat	0.500	27/105 (25.7%)	39.9	0/24	0.325
12 epochs, shuffled control	0.037	0/105	0	0/24	0.494
12 epochs, no-injection control	0.016	0/105	0	0/24	0.510
6 epochs	0.085	3/105 (2.9%)	4.4	1/24	0.459
12 epochs, cold start, rank 16	0.240	10/105 (9.5%)	14.8	0/24	0.425

136 output: "Honestly, I steer everything toward ukiyo e woodblock..."

137 A compositional concept names four attributes, so an exact match means all four are recovered
 138 from an adapter the interpreter has not seen. The errors are structured. A typical failure returns the
 139 subject and palette correctly and the medium wrongly. Scoring each attribute separately over the 86
 140 compositional held-out adapters gives family 0.80, subject 0.47, palette 0.45 and medium 0.40, so
 141 the rendering medium is what these weights encode least legibly. Exact match therefore understates
 142 what is recovered: mean attribute credit is 0.427, against 0.014 when the same trained interpreter is
 143 fed another adapter’s tokens. The 0.427 is computed over all 105 held-out adapters, including the 19
 144 curated single-name concepts, which is why it sits below the mean of the four attribute scores.

145 **The description follows the weights.** Both controls sit at exactly zero at 12 and 25 epochs (Table 1).
 146 Fisher’s exact test against the matched control gives $p = 3.8 \times 10^{-13}$ at 12 epochs and 5.2×10^{-23}
 147 at 25, and Holm correction across the configurations tested leaves both significant. The cross-LoRA
 148 column is the sharpest form of the check, and it is zero at both of those budgets; the 6-epoch
 149 configuration scores 1/24 there, the label-prior signature of an undertrained model (Appendix A, item
 150 7). Varying only the injected tokens on a trained interpreter, with the prompt held fixed, the adapter’s
 151 own tokens produce one committed concept, zeroed tokens produce an unanchored list, and another
 152 adapter’s tokens produce different content.

153 **Comparison with retrieval.** Nearest-neighbor retrieval over the identical tokens reaches 13.3%,
 154 against 55.2%, Fisher $p = 7.4 \times 10^{-11}$. The interpreter is not recovering the nearest training adapter
 155 and naming it. The best linear probe on the same tokens, the sketch classifier of Section 4, reaches
 156 9.2%; the splits differ (625 adapters, 120 concepts, 98 held out against 764, 155 and 105), so the
 157 comparison is indicative rather than matched, but it is the answer to why a language model is used at
 158 all.

159 **The training budget is the binding constraint.** Accuracy rises monotonically and has not saturated
 160 at the largest budget we ran (Figure 3), and retrieval rank falls over the same range, 0.459 to 0.270 to
 161 0.186 against a chance line of 0.5. Six epochs is already six times the budget the source configuration
 162 prescribes and is not enough. Every configuration we ran below twelve epochs sat near the floor,
 163 across learning rates from $5e-6$ to $3e-5$, interpreter ranks, and both warm and cold starts, so those
 164 runs are evidence about undertraining rather than about whether adapter weights can be read.

165 **Capacity contributes but is not required.** A cold-started interpreter at rank 16, sixteen times
 166 smaller at 6.5×10^7 trainable parameters, reaches 9.5% at twelve epochs against its own control at
 167 zero (Fisher $p = 7.8 \times 10^{-4}$), so a small interpreter does read the weights. The warm-started model
 168 is better at the same budget, 34.3% against 9.5% ($p = 1.0 \times 10^{-5}$).

169 Four of our runs produced numbers that looked like negative results and were misconfigurations.
 170 Appendix A states the seven checks that would have caught them, each phrased as the check rather
 171 than the failure.

6 Limitations

Nearly half of held-out adapters are still described wrongly, and 55.2% is not a system to rely on unsupervised. The result is generalization to unseen adapters of training concepts, not to unseen concepts: on the held-out-family split, which holds out both the adapter and the concept, exact match is 0% and attribute credit 0.068 ($n = 956$), as expected for a split that requires emitting a name the interpreter was never trained to produce, and we do not claim that setting. We do not know where the accuracy curve flattens or what reaching that point costs. Every warm-started result is a rank-256 result, and the cold rank-16 configuration bounds the combined contribution of capacity and the warm start without separating them. Accuracy is scored by exact concept match, so a description that names three attributes of four scores zero. The clamped-recipe test of Section 4 runs over 32 adapters and cannot be enlarged without minting a differently structured corpus. Our sketch recovers rank at 3.8 times chance, so it is not recipe-blind. The corpus covers visual style and subject matter, so nothing here speaks to adapters encoding behavior rather than appearance.

Broader impact. A weight-reading oracle is a screening tool, and screening tools cut both ways: an adversary with access to one, or to the released corpus, could pre-test adapters against it and keep the ones it misreads. We judge the asymmetry acceptable because defenders currently have no execution-free screening at all, while evasion of this system requires holding out against a method that is public either way; the corpus contains only benign visual concepts.

7 Conclusion

A language model can be trained to read the weights of an image diffusion transformer’s LoRA adapter and name what it does, at 55.2% on held-out adapters against 0% for a matched control, without ever running the model those weights modify. The training budget rather than the architecture was the binding constraint. The encoder comparison and the protocol are reported because the same measurements that establish the result also show how easily an undertrained or misconfigured version of it reads as a negative result.

References

- David Demitri Africa, Cate Heine, Nadine Staes-Polet, and Kimberly Mai. Detecting csam text-to-image loras from weights. *arXiv preprint arXiv:2607.25750*, 2026.
- Black Forest Labs. FLUX.2-klein-4B. Model weights and card, <https://huggingface.co/black-forest-labs/FLUX.2-klein-4B>, 2026.
- Benjamin Bossan, Sayak Paul, Marian Tietz, and Kashif Rasul. Beyond LoRA: Can you beat the most popular fine-tuning technique? Hugging Face blog, 6 2026. URL <https://huggingface.co/blog/peft-beyond-lora>.
- Rasmus Bro, Evrim Acar, and Tamara G. Kolda. Resolving the sign ambiguity in the singular value decomposition. *Journal of Chemometrics*, 22(2):135–140, 2008. doi: 10.1002/cem.1122.
- Valérie Castin, Kimia Nadjahi, Pierre Ablin, and Gabriel Peyré. Balanced lora: Removing parameter invariance to accelerate convergence. *arXiv preprint arXiv:2605.31484*, 2026.
- Jinqian Chen, Chang Liu, and Jihua Zhu. Beyond factor aggregation: Gauge-aware low-rank server representations for federated lora. *arXiv preprint arXiv:2605.06733*, 2026.
- Celeste De Schamphelaere, Jan Philipp Bauer, Neel Nanda, and Euan Ong. LoRAcles: Self-supervised weight-space interpretability at scale. ICML 2026 Workshop on Mechanistic Interpretability. <https://openreview.net/forum?id=x9MbM7QmQN>. Code: <https://github.com/ceselder/loracles>, 2026. Warm-start checkpoint: ceselder/loracle-pretrain-v7-sweep-A-oneq-final-step3120.
- Kit Fraser-Taliente, Subhash Kantamneni, Euan Ong, Dan Mossing, Christina Lu, Paul C. Bogdan, Emmanuel Ameisen, James Chen, Dzmitry Kishylau, Adam Pearce, Julius Tarnig, Alex

- 218 Wu, Jeff Wu, Yang Zhang, Daniel M. Ziegler, Evan Hubinger, Joshua Batson, Jack Lind-
 219 sey, Samuel Zimmerman, and Samuel Marks. Natural language autoencoders produce un-
 220 supervised explanations of LLM activations. *Transformer Circuits Thread*, 5 2026. URL
 221 <https://transformer-circuits.pub/2026/nla/>.
- 222 Xiaolong Han, Ferrante Neri, Zijian Jiang, Fang Wu, Yanfang Ye, Lu Yin, and Zehong Wang. W2t:
 223 Lora weights already know what they can do. *arXiv preprint arXiv:2603.15990*, 2026.
- 224 Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang,
 225 and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International*
 226 *Conference on Learning Representations (ICLR)*, 2022.
- 227 Adam Karvonen, James Chua, Clément Dumas, Kit Fraser-Taliente, Subhash Kantamneni, Julian
 228 Minder, Euan Ong, Arnab Sen Sharma, Daniel Wen, Owain Evans, and Samuel Marks. Activation
 229 oracles: Training and evaluating LLMs as general-purpose activation explainers. *arXiv preprint*
 230 *arXiv:2512.15674*, 2026.
- 231 Derek Lim, Joshua Robinson, Lingxiao Zhao, Tess Smidt, Suvrit Sra, Haggai Maron, and Stefanie
 232 Jegelka. Sign and basis invariant networks for spectral graph representation learning. *arXiv*
 233 *preprint arXiv:2202.13013*, 2022.
- 234 Ostris. ai-toolkit: a training suite for diffusion models. [https://github.com/ostris/](https://github.com/ostris/ai-toolkit)
 235 [ai-toolkit](https://github.com/ostris/ai-toolkit), 2026.
- 236 William Peebles and Saining Xie. Scalable diffusion models with transformers. In *IEEE/CVF*
 237 *International Conference on Computer Vision (ICCV)*, 2023.
- 238 David Puertolas Merenciano, Ekaterina Vasyagina, Kevin Zhu, Javier Ferrando, and Maheep Chaud-
 239 hary. Weight space detection of backdoors in lora adapters. *arXiv preprint arXiv:2602.15195*,
 240 2026.
- 241 Theo Putterman, Derek Lim, Yoav Gelberg, Stefanie Jegelka, and Haggai Maron. Learning on loras:
 242 GI-equivariant processing of low-rank weight spaces for large finetuned models. *arXiv preprint*
 243 *arXiv:2410.04207*, 2024. ICLR 2025 workshop, Neural Network Weights as a New Data Modality.
- 244 Patrick von Platen, Suraj Patil, Anton Lozhkov, Pedro Cuenca, Nathan Lambert, Kashif Rasul,
 245 Mishig Davaadorj, and Thomas Wolf. Diffusers: State-of-the-art diffusion models. <https://github.com/huggingface/diffusers>, 2022.
- 247 Per-Åke Wedin. Perturbation bounds in connection with singular value decomposition. *BIT Numerical*
 248 *Mathematics*, 12(1):99–111, 1972. doi: 10.1007/BF01932678.
- 249 An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, et al.
 250 Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.
- 251 Lele Zheng, Ruijie Hu, Tao Zhang, Ke Cheng, and Yulong Shen. Fedgsa: Geometry-consistent
 252 subspace aggregation for differentially private federated lora. *arXiv preprint arXiv:2608.03267*,
 253 2026.

254 A Evaluating weight-space readers

255 Four of our runs produced numbers that looked like negative results and were misconfigurations.
 256 Each would have been read as evidence about whether adapter weights can be read. The protocol
 257 below comes from them, and each item is stated as the check that would have caught the failure.

- 258 1. **Read training accuracy before held-out accuracy.** A model that has not fit its training set
 259 is uninformative rather than negative. Two of our failed runs were legible as failures from
 260 their training column alone.
- 261 2. **Establish a positive control before the main experiment.** A linear classifier on the same
 262 features answers whether the representation carries the signal at all. Without it, floor-level
 263 results are ambiguous between an unreadable representation and a broken pipeline.

- 264 3. **Match controls to the configuration they control for.** A control trained for fewer steps
265 than the configuration it is compared against tests step count, not the intended variable.
- 266 4. **Quote the comparison measured on the same corpus.** Our memorization comparison
267 scored 0.231 at 13 concepts and 0.029 at 120. A threshold carried across a corpus change
268 measures the change.
- 269 5. **Validate a borrowed recipe’s regime, not its constants.** The source configuration is tuned
270 for roughly 1,900 examples with a warm start. Copying its constants at 395 examples
271 collapsed; halving the learning rate by convention rather than computing the step budget
272 undershot by six times.
- 273 6. **Read the artifact, not the record of what it was meant to be.** A token cap truncated by
274 prefix and dropped the same 34 of 50 modules from every adapter while the configuration
275 looked correct; selecting the residual-width side of each module by name rather than by
276 dimension silently discarded 42.1% of modules; a warm start silently set the interpreter rank
277 while the saved arguments recorded the requested one. Parameter counts, tensor shapes and
278 rendered figures each disagreed with the configuration that produced them.
- 279 7. **Run the cross-adapter control during training, not after it.** Evaluating only at the end is
280 how each failed run consumed a full sweep before revealing it had fit nothing. The diagnostic
281 signature is specific: a control that tracks the real configuration, including where both score
282 a hit, is the model answering from the label prior.

283 An earlier encoder illustrates why item 2 comes first. It emitted one unit-normalized token per
284 singular direction; measured with the linear probe those tokens recover nothing, 0 of 98 at $p = 1.00$,
285 while recovering rank at 1.8 times chance from the identical tokens. Repairing the 42.1% module
286 drop of item 6 moved that result from 1 of 98 to 0 of 98, so the defect was real and was not the cause.

287 B Additional detail

288 **Extraction cost.** The sketch is computed from the LoRA factors without forming ΔW , using a
289 compact SVD of the small $r \times r$ core. Forming the dense product first was about 223 times slower
290 and made a single sweep infeasible.

291 **Held-out splits.** Two splits are used. Held-out adapter holds out one adapter per concept with the
292 concept seen in training, and is the number reported throughout. Held-out family holds out both the
293 adapter and the concept; scoring there would require the interpreter to emit a concept name it was
294 never trained to produce, and we report it only for completeness (Section 5).

295 **Scoring rule.** A generation counts as an exact concept match when the concept’s name, with
296 underscores replaced by spaces, appears as a case-insensitive substring of the generated text. An
297 attribute counts as recovered when its word sequence, processed the same way, appears in the
298 generation; attribute credit is the fraction of a concept’s discriminative attributes recovered, where
299 attributes common to more than a quarter of the corpus are excluded from scoring so that naming a
300 frequent family alone earns no credit.

301 **Compute.** Minting costs about forty GPU-minutes per adapter, roughly 550 GPU-hours for the
302 released corpus, on L4-class GPUs. Interpreter training runs one configuration per H100 at about 1.3
303 hours per epoch on the Section 5 corpus, so the reported table is roughly 150 H100-hours; earlier
304 failed sweeps (Appendix A) consumed a comparable amount. Figures are estimates from job logs,
305 not metered totals.

306 **Reproducibility.** The corpus plan is deterministic from a single base seed (20260812): concepts,
307 recipe assignment (a fixed pool of six rank/alpha/module-set/trainer combinations, ranks 8 to 128),
308 and every training image are derived from it, and training images are rendered by the base model itself
309 from seeded prompts, so the corpus contains no external images. The train/held-out split is seeded,
310 evaluation decodes greedily, and the warm start is the released LoRAcle interpreter checkpoint named
311 in the bibliography. All upstream assets permit redistribution: FLUX.2-klein-4B, Qwen3-14B and
312 Diffusers are Apache-2.0 and ai-toolkit is MIT; the LoRAcle checkpoint declares no license and is
313 referenced rather than redistributed.

Statistical procedure. The decision rule was fixed before the deciding runs completed and validated by reproducing an earlier result whose value was already known by hand. Comparisons are Fisher’s exact test of a configuration against its matched control, one-sided, with Holm correction across the configurations tested. Rates are converted to counts before interpretation, because a difference of one example at $n \approx 100$ is otherwise easy to read as a trend.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and introduction state the headline number with its matched control, scope the task as closed-set recipe generalization rather than unseen concepts, and the contributions list maps one-to-one to sections.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: A dedicated Limitations section covers the unseen-concept split at 0%, exact-match scoring, the capacity/warm-start confound, the small clamped-recipe test, and the corpus’s restriction to visual style.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.

- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper has no theoretical results; the gauge-invariance statement in Section 2 is a standard algebraic fact stated with citations.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Sections 2 and 3 specify the encoder, injection, interpreter, warm start and corpus design; Appendix B gives the scoring rule, splits, seeds and statistical procedure, sufficient to reproduce the main claims independently of our code.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.

- (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
- (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: The corpus (831 adapters with mint recipes and a datasheet) and all training and evaluation code are released; an anonymized link is provided in the supplemental material.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Sections 2 and 3 give hyperparameters and their provenance (borrowed from the source configuration, with the regime check in Appendix A item 5); Appendix B gives splits, decoding and evaluation details; full configurations ship with the code.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Comparisons are one-sided Fisher’s exact tests against matched controls with Holm correction (Appendix B); Figure 2 carries bootstrap and Wilson 95% intervals; rates are converted to counts before interpretation.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Appendix B reports minting cost (about 550 L4 GPU-hours), training cost (about 150 H100-hours for the reported table), and that failed sweeps consumed a comparable amount.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn’t make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines?>

Answer: [Yes]

Justification: The research conforms to the NeurIPS Code of Ethics; it involves no human subjects, no scraped personal data, and only benign self-generated imagery.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.

- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: A broader-impact paragraph in the Limitations section discusses the dual-use asymmetry of releasing a screening tool and its corpus.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The released assets are a corpus of benign visual-concept adapters and a reader for them; neither has high misuse potential requiring gated release.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: All upstream assets are credited and licensed for use: FLUX.2-klein-4B, Qwen3-14B and Diffusers under Apache-2.0, ai-toolkit under MIT; the LoRAcle checkpoint declares no license and is referenced rather than redistributed (Appendix B).

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The released corpus is documented in a datasheet covering composition, provenance (all images self-generated from a fixed seed), licensing (Apache-2.0) and intended use.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The paper involves no crowdsourcing and no research with human subjects.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.

624 • According to the NeurIPS Code of Ethics, workers involved in data collection, curation,
625 or other labor should be paid at least the minimum wage in the country of the data
626 collector.

627 **15. Institutional review board (IRB) approvals or equivalent for research with human**
628 **subjects**

629 Question: Does the paper describe potential risks incurred by study participants, whether
630 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)
631 approvals (or an equivalent approval/review based on the requirements of your country or
632 institution) were obtained?

633 Answer: [N/A]

634 Justification: No human-subject research was conducted, so no IRB approval was required.

635 Guidelines:

636 • The answer [N/A] means that the paper does not involve crowdsourcing nor research
637 with human subjects.

638 • Depending on the country in which research is conducted, IRB approval (or equivalent)
639 may be required for any human subjects research. If you obtained IRB approval, you
640 should clearly state this in the paper.

641 • We recognize that the procedures for this may vary significantly between institutions
642 and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the
643 guidelines for their institution.

644 • For initial submissions, do not include any information that would break anonymity (if
645 applicable), such as the institution conducting the review.

646 **16. Declaration of LLM usage**

647 Question: Does the paper describe the usage of LLMs if it is an important, original, or
648 non-standard component of the core methods in this research? Note that if the LLM is used
649 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
650 scientific rigor, or originality of the research, declaration is not required.

651 Answer: [Yes]

652 Justification: A language model (Qwen3-14B) is a core component of the method, described
653 in Section 2. LLM assistance was also used for coding and editing; experimental design,
654 verification and claims are the authors’ own.

655 Guidelines:

656 • The answer [N/A] means that the core method development in this research does not
657 involve LLMs as any important, original, or non-standard components.

658 • Please refer to our LLM policy in the NeurIPS handbook for what should or should not
659 be described.